A second-order cone representation of the angular determinant

23 October 2026

The angular determinant of a canonical Bell inequality contains eight nonconstant Laurent monomials. Their coefficients are nonnegative; only the constant can be negative. Positivity is therefore determined by how much constant each balanced group of exponents can support. There are twelve such elementary groups. We give their complete matrix representation, including vanishing coefficients and minima approached at infinity.

Put $E = \{-1, 0, 1\}^2 \setminus \{0\}$ and write $z^e = U^{e_1} V^{e_2}$. The cone to be represented is

$$\mathcal{L} = \{(h, q) : q_e \geq 0, \quad h + \sum_{e \in E} q_e z^e \geq 0 \quad (U, V > 0)\}. \tag{1}$$

Its nine coordinates are h and the eight q_e.

1 The twelve circuits and their matrices

Opposite pairs $\{e, -e\}$ give four circuits. Four more are $\{(s, t), (-s, 0), (0, -t)\}$, with $s, t \in \{-1, 1\}$. Their exponents add to zero. The remaining four have the form $\{e, f, g\}$, where e, f are adjacent corners of the square and $e + f + 2g = 0$. Denote the eight triangles by $\mathcal{T}$ and all twelve circuits by $\mathcal{C}$.

For each triangle C allocate three coefficients $r_{C,e}$ to its three monomials. The opposite pairs receive everything left over:

$$R_e = q_e - \sum_{C \in \mathcal{T} : e \in C} r_{C,e}. \tag{2}$$

Each circuit also receives a constant s_C. The matrices below ensure that its allocated monomials dominate s_C for every positive U, V.

For a pair $C = \{e, -e\}$ require

$$\begin{pmatrix} 2R_e & s_C \\ s_C & 2R_{-e} \end{pmatrix} \succeq 0. \tag{3}$$

For a triangle whose exponents add to zero, abbreviate its three allocations by r_1, r_2, r_3, put $t = s_C/3$, and require

$$\begin{pmatrix} r_1 & x \\ x & r_2 \end{pmatrix} \succeq 0, \qquad \begin{pmatrix} r_3 & y \\ y & t \end{pmatrix} \succeq 0, \qquad \begin{pmatrix} x & t \\ t & y \end{pmatrix} \succeq 0. \tag{4}$$

For $e + f + 2g = 0$, use r_1, r_2, r_3 in that order, put $t = s_C/4$, and require

$$\begin{pmatrix} r_1 & x \\ x & r_2 \end{pmatrix} \succeq 0, \qquad \begin{pmatrix} x & t \\ t & r_3/2 \end{pmatrix} \succeq 0. \tag{5}$$

Finally impose the single scalar inequality

$$h + \sum_{C \in \mathcal{C}} s_C \geq 0. \tag{6}$$

Every matrix entry is affine in the nine inputs and the auxiliary variables. The allocations and R_e are automatically nonnegative because they occur on matrix diagonals.

Theorem 1. *Equations* (2)–(6) *represent* $\mathcal{L}$ *exactly. Their block diagonal pencil has 24 blocks of order two and one scalar block, hence total order 49. It uses 48 auxiliary variables and rational coefficients.*

Proof. We first show that the matrices imply positivity. For an opposite pair, (3) gives $s_C \le 2\sqrt{R_e R_{-e}}$. The arithmetic–geometric mean inequality then gives $R_e z^e + R_{-e} z^{-e} \ge s_C$.

In (4), all diagonal entries are nonnegative and $x^2 \le r_1 r_2$, $y^2 \le r_3 t$, $t^2 \le xy$. If $t > 0$, these imply $t^3 \le r_1 r_2 r_3$; if $t = 0$ the desired bound is immediate. Since the three monomials have product one, their allocated sum is at least $3(r_1 r_2 r_3)^{1/3} \ge s_C$.

Similarly, (5) implies $t^4 \le r_1 r_2 (r_3/2)^2$. Applying the arithmetic–geometric mean inequality to $r_1 z^e, r_2 z^f, (r_3/2) z^g, (r_3/2) z^g$ shows that their sum is at least $4|t| \ge s_C$. Thus negative s_C causes no problem in either a pair or a weighted triangle; the ordinary triangle already forces $s_C \ge 0$. Summing all twelve bounds and using (2) and (6) proves membership in (1).

For the converse, suppose first that $h \ge 0$. Set all triangle allocations, constants and auxiliary variables to zero. The pair blocks are nonnegative diagonal matrices, and (6) holds. Now let $h < 0$ and let $I = \{e : q_e > 0\}$. Positivity forces $0 \in \operatorname{conv} I$: otherwise a separating direction in $(\log U, \log V)$ makes every active monomial tend to zero.

Let J be the active exponents on the smallest face of $\operatorname{conv} I$ containing zero. An exposing direction sends all off-face monomials to zero and leaves the face monomials fixed. Hence $h + \sum_{e \in J} q_e z^e$ is nonnegative. In the span of J, the exponential sum is strictly convex and tends to infinity along every unbounded ray. It has a positive minimizer $z_c = (U_c, V_c)$, unique modulo directions orthogonal to J. Put

$$w_e = q_e z_c^e \quad (e \in J), \qquad w_e = 0 \quad (e \notin J).$$

Stationarity and positivity at this minimum give

$$\sum_e w_e e = 0, \qquad h + \sum_e w_e \ge 0. \tag{7}$$

A nonnegative balanced vector on E decomposes into the twelve circuits above. Indeed, a minimal positive dependence uses an opposite pair or a triangle containing zero in its interior. The eight-point square has exactly the four equal-weight and four $(1,1,2)$ triangles listed above. Subtract the largest nonnegative multiple of an active minimal dependence and repeat. Each subtraction removes at least one positive coefficient, so there are at most eight steps. Write the result as $w = \sum_C \lambda_C b^C$, where $\lambda_C \ge 0$ and b^C has weights $(1,1)$, $(1,1,1)$ or $(1,1,2)$.

Allocate $\lambda_C b_e^C / z_c^e$ to circuit C and set $s_C = \lambda_C \sum_e b_e^C$. Each circuit attains equality at z_c, so it meets exactly the corresponding geometric-mean bound. For an ordinary triangle choose $x = \sqrt{r_1 r_2}$ and $y = \sqrt{r_3 t}$; the equality $t^3 = r_1 r_2 r_3$ makes all three blocks positive semidefinite. For a weighted triangle choose $x = \sqrt{r_1 r_2}$; its second block has determinant zero. The pair block also has determinant zero. Zero circuit weight gives zero blocks without any division by that weight.

The off-face coefficients have not been used. Assign each to its opposite-pair allocation; this only increases the corresponding diagonal entry. Thus the pair allocations are exactly (2). Equation (7) gives (6), proving the converse even when the original Laurent polynomial has no finite minimizer.

There are 24 triangle allocations, 12 constants s_C, eight variables for the four ordinary triangles and four for the weighted triangles: 48 auxiliaries in total. The pair, ordinary-triangle and weighted-triangle blocks number $4 + 12 + 8 = 24$. □

The accompanying `ANGULAR_CONE_ORDER49.coo.json` lists all 109 nonzero rational coefficient records of the pencil. Off-diagonal records are mirrored without a factor of one half. A standard-library checker reconstructs every coefficient matrix and checks exact boundary, zero-weight and off-face witnesses.

2 Application to canonical Bell inequalities

For nonnegative canonical coefficients $c = (d, \lambda, n, m, C, A, D, \alpha, \beta)$, the state determinant is

$$\det P_{\mathrm{aff}}(u, v) = 16(1+u^2)^2(1+v^2)^2u^2v^2\left(h(c) + \sum_{e\in E} q_e(c)u^{2e_1}v^{2e_2}\right).$$

The nine polynomials $h(c), q_e(c)$ are given in the accompanying determinant certificate and coefficient file. All outer coefficients are nonnegative on this orthant. If $d > 0$ and $uv \neq 0$, the three-dimensional transverse state block is positive definite. The Schur complement therefore shows that $P_{\mathrm{aff}} \succeq 0$ exactly when its determinant is nonnegative. Continuity includes the testing axes and projective endpoints. When $d = 0$, the full state polynomial is already a nonnegative sum of event squares.

Consequently a canonical Bell inequality is quantum-valid if and only if the order-49 pencil is feasible at $(h(c), q(c))$. This gives a smaller exact validity test than completing the entire state polynomial. The coefficient map is quartic in c. The pencil is affine in the nine Laurent coefficients; it is not an affine lift in the Bell coefficients or the eight quantum expectations. The separate order-133 construction supplies the fixed square factors for the determinant- weighted full state polynomial.